

CWU Design System

Cognitive Work Unit Composition Methodology

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What This Is

This document is the operating specification for how Polynize designs and deploys Cognitive Work Units. It serves two purposes simultaneously: a persistent intellectual artifact capturing the composition methodology, and a prompt engineering reference for building the CWU pre-mapper pipeline.

A Cognitive Work Unit (CWU) is a complete operating unit — not a role, not a job description — configured around one accountable human outcome owner, with agent layers carrying decomposable cognitive work and the human remaining central where trust, lived judgment, accountability, or high-stakes exceptions matter.

The core principle: **we are not composing teams from titles. We are composing Cognitive Work Units from atomic task-capability units, then clustering those into human and agent functions around one accountable human outcome owner.**

Part 1: Methodology

1.1 The Atomic Unit

The base primitive of CWU composition is not a role but a **task-capability pair under conditions**.

The minimum viable schema for v1:

Field	Definition
Task	What is being done
Capability	What skill or judgment is required to do it
CQ signal	Measured human capability strength (from Accelerator, where available)
Human Criticality	Low / Medium / High — how central must the human remain for this task?
Risk Sensitivity	Low / Medium / High — how dangerous is weak execution here?
Recommended Allocation	Human-led / Agent-assisted / Agent-executable / Human-review-required

Design rule: Do not add dimensions until they change allocation decisions. If a field does not materially affect who owns the task, how tightly it must be supervised, whether anti-drift is needed, or how the CWU is composed — leave it out.

1.2 Allocation Logic

Human Criticality drives the primary allocation decision. Risk Sensitivity drives the supervision and anti-drift design.

Human Criticality	Risk Sensitivity	Allocation
High	Any	Human-led
Medium	High	Human-review-required
Medium	Low/Medium	Agent-assisted
Low	High	Agent-executable + control layer
Low	Low	Agent-executable

What makes a task Human-led:

- Requires judgment built from high-stakes experience (not just skill)
- Requires genuine trust, relational presence, or accountability
- Carries high failure cost if executed wrongly
- Requires social nuance, commitment, or live decision-making
- Involves regulatory, fiduciary, or legal accountability

What makes a task Agent-executable:

- Structured, repeatable process with defined inputs and outputs
- Research, synthesis, drafting, formatting, coordination, tracking
- Pattern-matching against known corpora
- First-pass generation that a human reviews

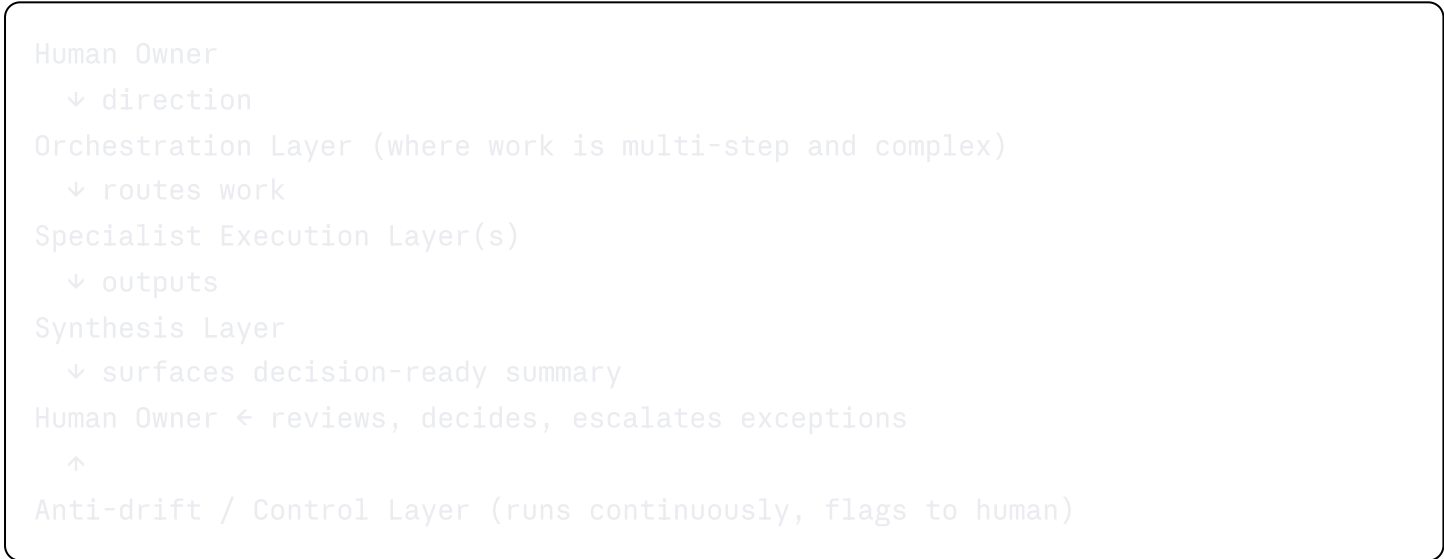
1.3 Functional Clusters

Once tasks are allocated, they cluster into four functional types. Every CWU contains some combination of these.

Cluster	What it does	Typical allocation
Framing	Deciding what matters, setting direction, prioritising	Human-led
Specialist Execution	Doing the domain work within defined scope	Agent-executable
Synthesis	Reconciling parallel outputs into a coherent decision surface	Agent-executable (output is human-reviewed)
Anti-drift / Control	Checking alignment, detecting contradictions, triggering escalation	Agent-executable

1.4 CWU Layer Architecture

Every CWU assembles into the same structural pattern, regardless of domain:



The five structural principles:

- One accountable human at the center.** Every CWU has a single human owner of the outcome. Not a team. Not a committee. One person.
- Compose by function, not title.** "Developer," "analyst," "sales rep" are too coarse. Design from actual work functions.
- Synthesis is a first-class function.** Parallel outputs do not become coherent on their own. Synthesis must be explicitly designed.
- Anti-drift must exist as a function.** Faster execution increases the cost of wrongness. Every leveraged unit needs alignment checking, contradiction detection, and coherence control.
- Measure at the outcome level.** Throughput quality, decision cycle time, strategic responsiveness — not seat utilisation or activity volume.

1.5 The Composition Method (Step by Step)

Step 1: Define the outcome What business result does this unit exist to produce? Be concrete. Not "manage clients" — "retain 95% of accounts above £50k ARR."

Step 2: Name the human outcome owner Who is the single accountable human? Defined by what they carry that cannot be delegated — the judgment call, the relationship, the architectural decision, the attention and direction.

Step 3: Decompose all work in the unit List every task required to deliver the outcome — across the human and all agent functions. Do not filter yet.

Step 4: Score each task Apply Human Criticality and Risk Sensitivity to each task. Assign Recommended Allocation.

Step 5: Cluster into functional streams Group allocated tasks into Framing, Specialist Execution, Synthesis, and Anti-drift clusters.

Step 6: Assemble the layer architecture Configure the unit: human owner → orchestration (if complex) → specialist layers → synthesis → anti-drift. Name the agent functions in each layer.

Step 7: Add escalation rules Specify what must be kicked back to the human or external expert. Every CWU needs explicit escalation triggers, especially for high-risk-sensitivity tasks.

Step 8: Define unit metrics How is this unit's performance measured at outcome level?

1.6 Role Naming Convention

Agent functions are named using the pattern: [Domain/Function] + [Work Type]

Stable work type vocabulary:

- **Specialist** — domain execution within defined scope
- **Analyst** — structured analysis and data work
- **Coordinator** — sequencing, state management, follow-through
- **Researcher** — information gathering, synthesis preparation
- **Reviewer** — quality checking, alignment verification
- **Synthesiser** — cross-stream reconciliation and integration
- **Director** — human center (always human, never agent)
- **Monitor** — ongoing signal watching and alert generation
- **Strategist** — option generation, scenario framing (agent-assisted, human-led)

Domain/function term is inferred from client context. Work type is drawn from the stable vocabulary above.

Part 2: Shape Library

A CWU Shape is a canonical structural prior — the skeleton of a known unit type. When composing a CWU from real client context, the pre-mapper recognises the shape first, then instantiates it with client-specific tasks, pain points, and language.

Eight canonical shapes cover the space of business knowledge work. Complex engagements may be compound CWUs — one shape feeding another (most commonly Shape 1 → Shape 3).

Shape 1: Analysis and Judgment

What it is: A unit that converts information into a recommendation or decision. Multiple parallel specialist research streams feed a synthesis layer, which surfaces a decision surface for the human. The human never touches raw inputs — they review synthesised output and make the call.

Human center type: Senior judgment holder. Investment lead, research director, senior advisor, strategy lead. Carries domain expertise and is accountable for recommendation quality.

Structural signature: Parallel streams → synthesis → human judgment call. High synthesis and anti-drift requirement.

Typical task-capability clusters:

- Domain research streams (market, financial, legal, technical, competitive)
- Data gathering and structuring per stream
- First-pass analysis per stream
- Cross-stream synthesis and reconciliation
- Red-flag identification and escalation
- Recommendation or memo drafting
- Human review and final judgment

Agent functions: Parallel specialist agents per research domain, Synthesis Agent, Contradiction Detector / QA Agent

Where synthesis lives: Always present, always critical. Parallel streams can contradict each other. Synthesis must reconcile before the human sees anything.

Anti-drift design: Contradiction detection across streams, source quality checking, framing alignment (is the analysis answering the right question?), red-flag escalation.

Human is irreplaceable for: Weighting evidence under uncertainty, making the call, knowing what matters in this specific context, accountability for the recommendation.

Leverage source: Parallelising research streams, compressing time from question to synthesised decision surface, concentrating human time on judgment not groundwork.

Leverage range: 3–6× depending on stream count and research volume.

First agent (try-before-you-buy): Research Specialist Agent. Takes a brief, returns structured research on one domain stream. Immediate time saving, zero dependency on other agents. Gap exposed: "this is useful but I need it across all the streams and reconciled."

Discriminating intake signals:

- "We spend too much time on groundwork before we can form a view"
- "Our senior people are doing research that shouldn't need them"
- "We review a lot of deals / candidates / opportunities and it takes too long"
- Work is primarily about forming a view on something

Domain examples: Investment DD, market research, legal review, strategic planning, procurement evaluation, FP&A, policy analysis, competitive intelligence, clinical review.

Shape 2: Pipeline and Conversion

What it is: A unit that moves opportunities through sequential stages toward a commitment. The human holds the relationship and closes. Agents carry every scaffolding function around the human's live interactions.

Human center type: Relationship holder and closer. Founder, AE, BD lead, partner, recruiter, fundraiser. Carries trust and is the face of the commitment.

Structural signature: Sequential stages with human at key gates. Work flows forward through a pipeline, not in parallel streams. Human is frontstage at trust moments, backstage everywhere else.

Typical task-capability clusters:

- Target identification and qualification
- Outreach and warming sequences
- Pre-meeting research and intelligence prep
- Meeting support (briefs, notes, real-time assistance)
- Proposal or solution shaping
- Objection handling preparation
- Close and commitment management
- Post-conversion handoff and documentation

Agent functions: Targeting and Qualification Specialist, Research and Intel Analyst, Outreach Coordinator, Proposal Specialist, Deal Tracker and Follow-through Coordinator

Where synthesis lives: Lighter than Shape 1. Synthesis function here is deal-state coherence — does everything the prospect has been told add up across touchpoints? Are there contradictions?

Anti-drift design: Pipeline state accuracy, next-step discipline, deal-going-cold detection, objection pattern recognition.

Human is irreplaceable for: Live trust moments, discovery nuance, reading the room, handling real objections, making and receiving commitment.

Leverage source: Eliminating prep and admin burden from human, increasing outreach volume without proportional human time, tightening follow-through, keeping the human present only at high-value touchpoints.

Leverage range: 2-4× on human time efficiency, higher on pipeline volume capacity.

First agent: Meeting Prep and Intelligence Agent. Produces a structured brief before every meeting — company context, contact background, recent news, likely objections, suggested angles. Useful in isolation. Gap exposed: "great prep but I still lose track of follow-through after."

Discriminating intake signals:

- "I spend too much time on admin around meetings"
- "We're not following up consistently"
- "We can't scale outreach without hiring"
- "I need to be in more conversations but there aren't enough hours"

Domain examples: B2B sales, fundraising, BD and partnerships, recruitment, grant applications, enterprise sales cycles.

Shape 3: Execution and Delivery

What it is: A unit that takes a specification and builds something. The human is the architect and director — sets intent, defines quality, handles exceptions. Agents carry building, testing, iteration, and structured production.

Human center type: Architect, technical director, product owner, creative producer, delivery lead. Carries the design judgment and is accountable for what gets shipped.

Structural signature: Spec → decompose → build → test → iterate → deliver. Orchestration layer is critical because work is multi-step, stateful, and has dependencies. Anti-drift is spec alignment, not cross-stream contradiction.

Typical task-capability clusters:

- Specification decomposition and planning
- Task sequencing and orchestration
- Building, coding, drafting, or production
- Testing and quality assurance
- Iteration on feedback

- Edge case and exception handling
- Integration, assembly, and deployment
- Human review and sign-off

Agent functions: Orchestrator Agent (decomposes and sequences), Builder Agents (one or more, parallel where work is decomposable), QA and Evaluator Agent, Integration Agent

Where synthesis lives: QA function — does output match spec? Are components coherent with each other? Is the build regressing against prior state?

Anti-drift design: Spec alignment checking (are we still building the right thing?), regression detection, technical debt accumulation monitoring, intent-to-output distance tracking.

Human is irreplaceable for: Architecture and trade-off decisions, exception handling on novel failures, final quality judgment, deciding what matters when spec and reality diverge.

Leverage source: Compressing build cycles, parallelising independent workstreams, eliminating human time on structured and repeatable work, concentrating human on design and exception handling.

Leverage range: 3-7× depending on decomposability of the work.

First agent: Orchestrator / Spec Decomposition Agent. Takes a brief or spec and returns a decomposed, sequenced task list with dependencies. Removes the planning burden immediately. Gap exposed: "great plan but I still have to execute all of it myself."

Discriminating intake signals:

- "We can't ship fast enough"
- "Too much time on repetitive or boilerplate work"
- "We need to produce more output than our team can handle"
- Work is primarily about producing something to a specification

Domain examples: Software engineering, content production, marketing campaign execution, product development, financial reporting, data engineering, event production, document production at scale.

Shape 4: Executive Leverage

What it is: A unit built around one high-value human whose attention is the scarce resource. Agents expand the human's reach, reduce coordination loss, and ensure the human's time is concentrated on decisions only they can make.

Human center type: Founder, CEO, senior partner, department head, board-level executive. The human's judgment is unusually high-value. Time is the binding constraint.

Structural signature: Not a single output unit — continuous leverage across multiple parallel workstreams. The design problem is protecting the human's attention frontier and reducing translation loss between intent and execution.

Typical task-capability clusters:

- Inbox and communication triage
- Meeting preparation and briefing
- Research on demand
- Internal coordination and follow-through
- External outreach and relationship warming
- Strategic context maintenance
- Decision support on incoming requests
- Output drafting (communications, decks, documents)

Agent functions: Clearing and Triage Agent, Research and Intelligence Agent, Outreach and Warming Agent, Internal Coordination Agent, Drafting and Synthesis Agent

Where synthesis lives: Context harmonisation — the human receives synthesised summaries and prioritised decision queues, not raw inputs. Agents maintain a coherent picture of what is in flight across all workstreams.

Anti-drift design: Priority drift (urgent crowding out important), context fragmentation (human losing thread across parallel workstreams), follow-through gaps and commitment tracking.

Human is irreplaceable for: Strategic priority calls, relationship-level decisions, anything requiring the human's specific authority, trust, or vision.

Leverage source: Protecting the attention frontier, reducing time from decision to action, eliminating context-switching cost, compressing translation loss between intent and execution.

Leverage range: 2-4× on decision throughput, higher on reach and output volume.

First agent: Clearing and Triage Agent. Triages communications, surfaces what actually needs the human's attention, filters the rest. Useful from day one. Gap exposed: "I can see what matters now but I still have to do all the follow-through myself."

Discriminating intake signals:

- "I'm the bottleneck in my own organisation"
- "Too much of my time goes on things that shouldn't need me"
- "I lose track of what's in flight"
- "There's too much to coordinate and things are falling through"

Domain examples: Founder operations, CEO office, senior partner leverage, chief of staff function, department head with wide span of control.

Shape 5: Relationship Continuity

What it is: A unit that maintains ongoing relationships at scale — clients, accounts, partners. There is no pipeline end state. The work is continuous. The human holds relationship depth and handles anything requiring genuine trust or judgment. Agents carry monitoring, preparation, routine communication, and follow-through.

Human center type: Account manager, customer success lead, relationship partner, ongoing advisor. Carries relationship history and trust. The human's presence is the product.

Structural signature: Continuous, not project-based. No close event. Agents are always running in the background, surfacing signals. Human engages when the signal warrants it.

Typical task-capability clusters:

- Account health monitoring across signals
- Usage and engagement tracking
- Proactive outreach preparation
- Periodic review and QBR preparation
- Issue detection and early triage
- Escalation handling
- Renewal and expansion opportunity identification
- Routine communication and check-in management

Agent functions: Account Health Monitor, Engagement Signal Analyst, Review Prep Specialist, Communication Coordinator, Opportunity Detection Agent

Where synthesis lives: Account state synthesis — a coherent current picture of each relationship surfaced to the human before interactions. Not raw signal feeds but actionable account summaries.

Anti-drift design: Relationship neglect detection (accounts going quiet), signal-to-noise filtering, consistency across touchpoints, renewal risk early warning.

Human is irreplaceable for: Conversations requiring genuine empathy or trust, handling dissatisfaction or conflict, renewal negotiations, any moment where the human needs to be present as a person.

Leverage source: Scaling relationship coverage without proportional headcount, ensuring no account goes dark, concentrating human time on high-value interactions.

Leverage range: 2-4× on accounts managed per human, potentially higher on at-risk account recovery rate.

First agent: Account Health Monitor. Tracks engagement signals across accounts and surfaces at-risk relationships before they go cold. Standalone value. Gap exposed: "I can see the problem

but I need prep materials to actually act on it."

Discriminating intake signals:

- "We can't keep up with all our accounts"
- "We only talk to clients when they reach out to us"
- "We're losing accounts we didn't see coming"
- "Our CS team is stretched too thin"

Domain examples: Customer success, account management, ongoing advisory relationships, partner management, key account programmes.

Shape 6: High-Volume Operations with Exception Escalation

What it is: An inverted structure. Agents are the primary execution layer handling high-volume, structured work. The human is not the center of the work — the human is the exception handler and quality arbiter. The design problem is the opposite of most CWUs: protect the process from edge cases, not the human from routine work.

Human center type: Operations lead, compliance officer, service manager, quality lead. Carries exception judgment and is accountable for output quality at scale.

Structural signature: Inverted. Agents handle the majority. Human handles the exceptions. The synthesis function here is pattern reporting — what exceptions are occurring and what do they indicate about process quality or input changes?

Typical task-capability clusters:

- Intake classification of incoming work
- Structured processing (applications, claims, tickets, requests, monitoring)
- Standard response or output generation
- Quality sampling and audit
- Exception identification and routing
- Complex case handling
- Escalation to external expert where required
- Reporting and pattern detection

Agent functions: Classification and Routing Agent, Processing Agents (high volume, potentially parallel), Quality Sampling Agent, Exception Detection Agent, Pattern Reporting Agent

Where synthesis lives: Pattern reporting — what exception types are occurring, at what rate, what does that suggest about process quality or input signal changes? Surfaced to the human for process redesign decisions.

Anti-drift design: Exception rate monitoring (rising exceptions = process drift or input shift), quality degradation detection, compliance boundary checking, escalation threshold tuning.

Human is irreplaceable for: True exceptions outside defined scope, regulatory and legal judgment calls, process redesign when patterns indicate systematic failure, accountability for output quality at scale.

Leverage source: Handling volume impossible with human-only processing, concentrating human time on genuinely complex cases, improving consistency of standard processing, eliminating fatigue-driven error in repetitive work.

Leverage range: 5-20× on volume throughput — the highest raw leverage of any shape.

First agent: Classification and Routing Agent. Assesses every incoming item and routes it correctly. Standalone value at any volume. Gap exposed: "things are routed but the processing is still manual."

Discriminating intake signals:

- "We can't handle the volume"
- "Too much time on routine processing that feels like it shouldn't need a human"
- "We need to respond faster but we're understaffed"
- "Quality is inconsistent because the team is overwhelmed"

Domain examples: Customer support at scale, compliance monitoring, claims processing, loan or application processing, HR operations, content moderation, regulatory reporting.

Shape 7: Creative Direction

What it is: A unit where the human's taste and judgment is the generative source, not the review layer. The human sets creative direction, makes aesthetic and strategic calls, and defines what good looks like. Agents produce variants, handle production, and explore option spaces. The flow is: human direction → agent generation → human curation → agent production.

Human center type: Creative director, brand strategist, editor, designer, content lead. Carries taste, voice, and creative vision. Cannot be separated from the output's identity.

Structural signature: Human direction → agent option generation → human curation → agent production. The synthesis function here is option surface presentation — agents generate range for human curation, not a single output. Human resolves the option space.

Typical task-capability clusters:

- Creative brief development and enrichment
- Research and reference gathering
- Concept and variant generation
- Option space exploration

- Human curation and direction
- Production and refinement
- Adaptation across formats and channels
- Brand and voice consistency checking

Agent functions: Research and Reference Agent, Generation Specialists (copy, visual concept, structural), Production Agent, Adaptation and Formatting Agent, Consistency Checker

Where synthesis lives: Option surface — presenting the generated option space clearly for human curation. Not resolving it. The human resolves it.

Anti-drift design: Brand and voice consistency checking, creative drift detection across iteration cycles, quality regression monitoring across production volume.

Human is irreplaceable for: The taste call, the creative direction decision, the judgment of what is on-brand, the vision that gives the work coherence and identity.

Leverage source: Expanding what the creative director can produce without expanding headcount, compressing production cycles, enabling broader option exploration before committing to direction.

Leverage range: 2-5× on output volume, potentially higher on exploration breadth before decision.

First agent: Research and Reference Agent. Ingests a brief and returns a structured reference pack — competitive examples, relevant trends, audience signals, inspiration directions. Removes real creative friction. Gap exposed: "great inputs, but I'm still doing all the generation and production myself."

Discriminating intake signals:

- "We can't produce content at the quality and volume we need"
- "Our creative team is a bottleneck"
- "We want to explore more options before committing"
- "Production is taking too long after the creative direction is set"

Domain examples: Brand strategy, content production at scale, editorial direction, design systems, marketing creative, campaign development.

Shape 8: Learning and Capability Development

What it is: A unit designed to develop human capability at scale. The human expert or L&D lead defines what good looks like and holds the curriculum judgment. Agents deliver, assess, personalise, and surface cohort patterns. This is the shape that Polynize's Accelerator product embodies.

Human center type: L&D lead, subject matter expert, team leader, CHRO. Carries the definition of capability and the judgment about what the organisation needs to develop.

Structural signature: Capability assessment → individual gap identification → personalised development → cohort pattern synthesis → leadership decision support. The synthesis function is cohort intelligence — individual data aggregated into team-level patterns and action priorities.

Typical task-capability clusters:

- Capability needs analysis
- Curriculum and scenario design
- Content and scenario delivery at scale
- Individual assessment and CQ scoring
- Gap identification per individual
- Cohort pattern synthesis
- Personalised development pathway design
- Management reporting and visibility
- Remediation and follow-through tracking

Agent functions: Delivery Agent (simulations, scenarios), Assessment Agent (CQ scoring, gap analysis), Personalisation Agent (adaptive pathways), Pattern Synthesis Agent (cohort intelligence), Reporting and Visibility Agent

Where synthesis lives: Cohort intelligence — individual assessment data synthesised into team-level patterns, gap maps, and actionable development priorities for the human leader.

Anti-drift design: Assessment validity checking (are scores reflecting real capability or test-taking behaviour?), curriculum drift detection (does content still reflect what good looks like?), engagement signal monitoring, development pathway effectiveness tracking.

Human is irreplaceable for: Defining what good looks like, designing scenarios that reflect real work, interpreting cohort patterns in business context, making development investment and promotion decisions.

Leverage source: Delivering at scale without proportional expert time, generating individual gap maps previously requiring manual assessment, surfacing cohort patterns previously invisible, compressing development cycle time.

Leverage range: 5-15× on capability assessment coverage, 3-5× on development cycle time.

First agent: Assessment and Gap Analysis Agent. Delivers capability scenarios, scores performance, returns individual gap maps. This is the Accelerator's core. Gap exposed: "I can see the gaps but I don't have a systematic way to close them."

Discriminating intake signals:

- "We don't have visibility into our team's real capability"

- "Training isn't sticking — we don't know if it's working"
- "We can't assess at the scale we need"
- "We need to onboard faster"
- "We want to know who's actually ready before we promote"

Domain examples: Corporate L&D, onboarding programmes, sales capability development, leadership development, compliance training, capability assessment for workforce planning.

Part 3: Shape Coverage Reference

Shape	Primary Structure	Covers
1 · Analysis and Judgment	Parallel streams → synthesis → human call	Investment DD, market research, legal review, strategy, procurement, FP&A, competitive intel
2 · Pipeline and Conversion	Sequential stages, human at gates	Sales, BD, fundraising, recruitment, partnerships, grant applications
3 · Execution and Delivery	Spec → orchestrate → build → QA → ship	Software, content production, reporting, data engineering, campaigns, event production
4 · Executive Leverage	Continuous parallel workstreams, attention protection	Founder ops, CEO office, senior partner, chief of staff, department head
5 · Relationship Continuity	Continuous monitoring, human at high-value moments	Customer success, account management, ongoing advisory, partner management
6 · High-Volume Operations	Inverted — agents primary, human handles exceptions	Support at scale, compliance, claims, applications, content moderation
7 · Creative Direction	Human direction → agent generation → human curation → production	Brand, editorial, content strategy, design, campaign creative
8 · Learning and Capability	Assessment → gaps → development → cohort intelligence	L&D, onboarding, capability assessment, leadership development

Compound CWUs: Complex engagements often combine two shapes. The most common compound patterns:

- **Shape 1 → Shape 3:** Research and strategy phase feeding an execution and delivery phase
- **Shape 4 + Shape 2:** Executive leverage unit with an embedded pipeline function
- **Shape 1 + Shape 8:** Analysis of capability gaps feeding a development programme

Part 4: Pre-Mapper Specification

4.1 What the Pre-Mapper Does

The pre-mapper is a **first-pass CWU generator** that takes a meeting artifact or intake brief and produces a proposal-ready CWU design. It is not the full methodology — it uses the methodology implicitly and outputs at a higher level of abstraction.

Design brief: Generate a plausible first-pass CWU design, not a perfect final architecture. Optimise for credibility, clarity, and strategic leverage from incomplete input.

What it tolerates: Slightly messy input, incomplete workflow descriptions, vague pain points, approximate role descriptions.

What it must produce: A confident, coherent CWU concept that a prospect will find recognisable and compelling. Not a hedge. Not a vague framework. A specific design with a named human owner, named agent functions, a clear operating flow, and a directional leverage estimate with rationale.

4.2 Internal Inference Logic

When given a meeting artifact or intake brief, the pre-mapper runs this inference sequence:

1. **Extract the real outcome.** Not the stated problem — the business result the client actually needs. These are often different. "We're drowning in reporting" → outcome is probably "leadership makes timely decisions with accurate portfolio visibility."
2. **Recognise the shape.** Which of the eight canonical shapes does this most closely match? Commit to a shape. Note the assumption if input is ambiguous.
3. **Identify the human center by elimination.** Who in this workflow carries trust, accountability, and non-substitutable judgment? Usually clear from the meeting description.
4. **Decompose workstreams from the workflow description.** Even vague descriptions contain implicit stream boundaries. Name them using the role naming convention.
5. **Assign agent roles to repeatable and structured streams.** Research, analysis, drafting, scheduling, tracking, coordination are almost always agent-allocable. Be confident about this.
6. **Insert synthesis and anti-drift layer.** Clients almost never mention the coherence problem — they describe the work but not the integration requirement. Always add this and explain why.
7. **Estimate leverage directionally using the source of compression.** Name what is being compressed: coordination overhead, execution time, research burden, parallel stream capacity, follow-through consistency. Use bands not false precision.

8. **Surface assumptions and upgrade path.** Name what would need deeper mapping to validate the design.

4.3 Output Template

- 1. **CWU Name and Summary** A one-line name for the unit and a concise (2-3 sentence) description of what it is and what it produces.
- 2. **Human Owner** Who remains central and why — what they carry that cannot be delegated.
- 3. **Unit Design** 3-6 named functions with a one-sentence description of each and why it exists. Use the role naming convention.
- 4. **Operating Flow** Simple sequence: human direction → specialist work → synthesis / checking → human decision / action. Name the handoff points.
- 5. **Expected Leverage** Directional estimate (use bands: 1.5-2×, 2-4×, 3-5×, 5×+) with explicit rationale. Name the source: reduced coordination loss, parallel specialist streams, compressed prep and admin, tighter follow-through, concentrated human judgment.
- 6. **Assumptions and Open Questions** What input was incomplete. What would need validation before full deployment.
- 7. **Next-Step Upgrade Path** How this first-pass design becomes a fully mapped and deployed CWU — typically via Accelerator capability mapping, full task-capability decomposition, and pilot deployment.

4.4 Leverage Estimation Bands

Band	Range	Source
Low	1.5-2×	Primarily admin reduction, modest parallelisation
Moderate	2-4×	Parallel specialist work, prep and coordination elimination
High	3-5×	Full specialist parallelism, synthesis compression, human time concentrated on judgment only
Very high	5×+	High-volume operations, assessment at scale, or high decomposability

Framing for pre-sales use: "A unit shaped like this would likely create 2-4× leverage by [specific source], concentrating [human owner]'s time on [high-value activity] rather than [current time sink]." Never state a single precise number without empirical grounding.

Part 5: First-Agent Strategy

The first agent is the try-before-you-buy entry point. Design principle: **the one agent that**

removes the most friction from the human's current day without requiring the rest of the unit to exist first. It must deliver standalone value and expose a gap that makes the next agent the natural pull.

Shape	First Agent	What it removes	Gap it exposes
1 · Analysis and Judgment	Research Specialist	Human groundwork burden	"Useful but needs synthesis across all streams"
2 · Pipeline and Conversion	Meeting Prep and Intelligence Agent	Prep and admin around meetings	"Better prep but still losing follow-through after"
3 · Execution and Delivery	Orchestrator / Spec Decomposition Agent	Planning and decomposition burden	"Great plan but still executing everything manually"
4 · Executive Leverage	Clearing and Triage Agent	Communication overload and noise	"I can see what matters but still doing all the follow-through"
5 · Relationship Continuity	Account Health Monitor	Blind spots and accounts going dark	"I can see the problem but need prep materials to act"
6 · High-Volume Operations	Classification and Routing Agent	Triage and routing overhead	"Things are routed but processing is still manual"
7 · Creative Direction	Research and Reference Agent	Pre-creative research and reference gathering	"Great inputs but still doing all generation and production"
8 · Learning and Capability	Assessment and Gap Analysis Agent	Invisible capability gaps	"I can see the gaps but no systematic way to close them"

The pattern: The first agent is almost always either the research/intelligence layer, the intake/classification layer, or the monitoring/visibility layer. Never the synthesis layer or anti-drift layer — those require other agents to exist first.

The expansion motion: The gap the first agent exposes is the natural next purchase. The first agent is not just a product — it is the demonstration of the model and the case for completing the unit.

Part 6: Using This Document as a Prompt Spec

When using this document as context for a prompt that generates first-pass CWU designs, instruct the model to:

1. Accept intake context in natural language — meeting summaries, workflow descriptions, pain point statements, role descriptions
2. Run the internal inference logic from Part 4.2 without exposing it to the user
3. Recognise the canonical shape from Part 2 and name it
4. Instantiate the shape's structure with client-specific function names using the naming convention from Part 1.6
5. Apply the output template from Part 4.3
6. Estimate leverage using bands from Part 4.4 with explicit rationale — never a single number without empirical grounding
7. Commit to a design rather than hedging — name assumptions explicitly rather than producing a vague output

What the prompt should not do: Run the full atomic unit scoring methodology at the pre-mapper stage. Apply rigid templates that cannot flex to client language. Produce hedge language like "it depends" without following it with a specific answer. Overclaim precision on leverage estimates.

CWU Design System v0.1 · Polynize · April 2026 Internal working document — methodology and prompt engineering reference Next version: empirical calibration of leverage bands from pilot deployments, expanded domain-specific shape variants, full prompt pipeline specification